

ABHAY ZALA

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EXPERIENCE

HeyGen Technology Inc.

Sep 2024 – Present

Research Engineer

- **Multi-Modal Fine-Tuning:** Led fine-tuning projects for vision-language models and image diffusion models using PEFT/LoRA techniques to advance multi-modal understanding and improve facial similarity in generated images
- **Facial Enhancement & Aesthetics:** Developed computer vision pipelines for facial image enhancement and restoration, directly elevating digital twin visual fidelity and aesthetic quality
- **Model Evaluation & Benchmarking:** Led the development and training of specialized vision models to quantitatively evaluate and benchmark generated video quality across multiple perceptual axes
- **Inference Optimization:** Optimized end-to-end multi-modal video generation pipelines, drastically reducing model latency and compute overhead for production-scale deployment
- **Enterprise Agentic Harness:** Designed and deployed an enterprise-wide agentic harness for LLMs—now adopted globally across all product-facing agents at HeyGen—optimizing LLM execution speed, token efficiency, and operational costs
- **Open-Source Innovation:** Spearheaded the development of a code-as-video framework (<https://github.com/heygen-com/hyperframes>), a now open-source framework enabling LLMs to programmatically render and edit videos
- **Full-Stack ML Execution:** Owned end-to-end delivery of complex AI features spanning model research and development, backend services, and frontend UI design
- **Cross-Functional Team Leadership:** Served as Innovation Lead, driving R&D and product integration; initiated and led cross-functional teams to ship end-to-end automated video creation products
- **Mentorship & Talent Development:** Managed and mentored research engineering interns on the Innovation Team, guiding technical execution and model evaluation pipelines
- Two time “Gennovator” Award Winner (top performing engineer)

University of North Carolina at Chapel Hill

Sep 2019 – Sep 2024

Research Assistant, Supervised by Prof. Mohit Bansal

- Published several first- & second-author research papers and presented at prestigious conferences such as NeurIPS, CVPR, ICCV, NAACL, AAAI, and EMNLP
- Expertise in text-to-image generation and interpretable evaluation, text-to-video generation, visual programming with Large Language Models (LLMs), visual commonsense reasoning, vision-language navigation, image/video captioning, 3D pose understanding, PyTorch deep learning framework
- Serving as an organizer for the ACL Year-Round Mentorship program
- Mentored undergraduate research students

EngageAI

Mar 2023 – Sep 2024

Researcher

- Led a team of academic researchers from multiple universities on video processing and understanding for classroom settings
- Presented as an invited speaker to a large audience on the topics of research in practice, deep learning, and LLMs

Meta AI Research Collaboration

Feb 2022 – Mar 2023

Primary Researcher

- Led project development on video/moment retrieval, segmentation, and captioning in collaboration with researchers from Meta AI
- Published and presented a research paper at CVPR 2023

Capital One

Jun 2022 – Aug 2022

Machine Learning Intern

- Developed an internal document retrieval application
- Demonstrated expertise in document search (e.g., TF-IDF), vector database development, and feature extraction

EDUCATION

MS University of North Carolina at Chapel Hill
Major: Computer Science (Multimodal AI)

May 2022 – May 2024

BS University of North Carolina at Chapel Hill
Major: Computer Science

Aug 2019 – May 2022

Outstanding Undergraduate Researcher Award, Honorable Mention
Computing Research Association (CRA)

PUBLICATIONS

* Indicates equal contribution

10. Han Lin*, Jaemin Cho*, Abhay Zala, Mohit Bansal. **Ctrl-Adapter: An Efficient and Versatile Framework for Adapting Diverse Controls to Any Diffusion Model**. Proceedings of ICLR 2025 (Oral) [[pdf](#)]
9. Abhay Zala*, Jaemin Cho*, Han Lin, Jaehong Yoon, Mohit Bansal. **EnvGen: Generating and Adapting Environments via LLMs for Training Embodied Agents**. Proceedings of COLM 2024 [[pdf](#)]
8. Abhay Zala, Han Lin, Jaemin Cho, Mohit Bansal. **DiagrammerGPT: Generating Open-Domain, Open-Platform Diagrams via LLM Planning**. Proceedings of COLM 2024 [[pdf](#)]

7. Han Lin, [Abhay Zala](#), Jaemin Cho, Mohit Bansal. **VideoDirectorGPT: Consistent Multi-scene Video Generation via LLM-Guided Planning**. Proceedings of COLM 2024 [[pdf](#)]
6. Jaemin Cho, [Abhay Zala](#), Mohit Bansal. **Visual Programming for Text-to-Image Generation and Evaluation**. Proceedings of NeurIPS 2023 [[pdf](#)]
5. [Abhay Zala*](#), Jaemin Cho*, Satwik Kottur, Xilun Chen, Barlas Oğuz, Yasher Mehdad, Mohit Bansal. **Hierarchical Video-Moment Retrieval and Step-Captioning**. Proceedings of CVPR 2023 [[pdf](#)]
4. Jaemin Cho, [Abhay Zala](#), Mohit Bansal. **DALL-Eval: Probing the Reasoning Skills and Social Biases of Text-to-Image Generative Models**. Proceedings of ICCV 2023 [[pdf](#)]
3. Hyounghun Kim*, [Abhay Zala*](#), Mohit Bansal. **CoSim: Commonsense Reasoning for Counterfactual Scene Imagination**. Proceedings of NAACL 2022 [[pdf](#)]
2. Hyounghun Kim*, [Abhay Zala*](#), Graham Burri, Mohit Bansal. **FixMyPose: Pose Correctional Captioning and Retrieval**. Proceedings of AAI 2021 [[pdf](#)]
1. Hyounghun Kim, [Abhay Zala](#), Graham Burri, Hao Tan, Mohit Bansal. **ArraMon: A Joint Navigation-Assembly Instruction Interpretation Task in Dynamic Environments**. Findings of EMNLP 2020 [[pdf](#)]

TALKS, PRESENTATIONS, AND WORKSHOPS

2023 NeurIPS Paper Presentation, “Visual Programming for Text-to-Image Generation and Evaluation”

2023 EngageAI Panel, “Research in Practice”

2023 EngageAI Talk, “Deep Learning and Large Language Models”

2023 CVPR Paper Presentation, “Hierarchical Video-Moment Retrieval and Step-Captioning”

2023 EngageAI Research Presentation, “Video Moment-Retrieval and Moment-Captioning on Classroom Videos”

2022 NAACL Paper Presentation, “CoSim: Commonsense Reasoning for Counterfactual Scene Imagination”

2021 AAI Paper Presentation, “FixMyPose: Pose Correctional Captioning and Retrieval”

2020 SpLU Workshop, “ArraMon: A Joint Navigation-Assembly Instruction Interpretation Task in Dynamic Environments”

SKILLS

Programming Languages: Python, Java, C#, C/C++, JavaScript, OpenGL, Svelte, NodeJS, LaTeX, HTML/CSS, PHP, SQL, MATLAB

Deep Learning Frameworks: PyTorch

Platforms: Unity Engine, Amazon Mechanical Turk, Docker, Kubernetes, Git, GitHub, Amazon Web Services (AWS), Google Cloud Platform (GCP), IBM Cloud, Linux, Windows, MacOS, Google Firebase, Adobe Suite, Autodesk Suite, Microsoft Office Suite, Overleaf

Applications: Deep Learning, Machine Learning, Natural Language Processing (NLP), Computer Vision, Robotics, Multimodal AI, Dataset Creation, Simulator Development, Website Development, Software Development, Database Management, VR Development, Graphics Rendering

OTHER

- Red team member for OpenAI's DALL-E 2 [[information](#)]
- Site Developer/Maintainer of [perfect-type.com](#)
- ACL Year-Round Mentorship Logo Designer
- Developed Social Media Platform for Networking
- Developed Online Peer Tutoring Service with AI Assistance
- Developed TV Show and Movie Showcase Website
- Developed VR Healthcare Training System
- Developed an AI assistant for presentations
- Winner of several hackathons